

Edge-Aware Deep Learning for Robust American Sign Language Alphabet Classification

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Abstract—American Sign Language (ASL) recognition plays a vital role in enabling inclusive communication for the Deaf and hard-of-hearing community. However, existing deep learning approaches often rely on computationally heavy models, limiting deployment on mobile and edge devices. This paper presents a lightweight deep learning framework for ASL alphabet recognition using the MobileNetV3-Large architecture, fine-tuned through transfer learning on the publicly available ASL Alphabet dataset containing 87,000 images across 29 classes. Preprocessing and augmentation were applied to improve robustness against variations in lighting, background, and signer differences. The final model achieved a test accuracy of 85.71%, with most classes attaining high precision and recall, though misclassifications occurred for visually similar gestures such as “M” vs. “N” and “S” vs. “T”. Performance analysis using accuracy curves, confusion matrices, and F1-scores confirmed the model’s ability to generalize effectively. The lightweight nature of MobileNetV3 enables low-latency inference, making the system suitable for resource-limited environments such as smartphones and embedded devices. This work contributes an efficient, deployable solution for ASL recognition, with future directions including augmentation for signer diversity, real-time webcam integration, and coupling with natural language processing to support continuous sign-to-speech translation.

Index Terms—Sign Language Recognition, Deep Learning, American Sign Language, MobileNet, Computer Vision

I. INTRODUCTION

Recognizing American Sign Language (ASL) is a practical, interesting, and socially meaningful computer vision task,

which has several practical applications and which can help reduce everyday communication barriers for the challenged Deaf and hard-of-hearing individuals, particularly when implemented as an assistive mobile application. Recent studies on ASL recognition conducted in recent times have shown that deep convolutional neural networks (CNNs) can achieve strong performance at the alphabet level. However, it is to be noted that real-world accuracy is often affected by illumination changes, complex or cluttered backgrounds, variations in hand scale, and differences between signers [1] among various other challenges. Therefore, several previous studies have shown that as a result, recent research has emphasized specifically on lightweight backbone architectures and efficiency-aware training strategies instead of focusing only on maximizing accuracy [2]. In this specific setting, it is noted that transfer learning has emerged as a practical and widely adopted solution to obtain competitive performance from compact models. It has been successfully applied in real-time webcam-based recognition systems and deployment-oriented user interfaces among many other variations and technical settings [3]. Similarly, it can be shown that the models from the EfficientNet family have demonstrated that careful fine-tuning can deliver robust gesture recognition while remaining suitable for edge deployment, highlighting the importance of compact yet expressive feature extractors [4]. From a broader systems perspective, edge-cloud considerations such as latency, bandwidth usage, and scheduling constraints further support the need for efficient on-device inference to maintain responsiveness and scalability in practical environments [5]. Motivated by these techni-

cal and deployment-related requirements, our study focuses on lightweight transfer learning combined with robustness-oriented preprocessing as core components for practical and deployable ASL alphabet recognition.

A key challenge in ASL alphabet classification is the frequent confusion between visually similar hand shapes, particularly when small differences in finger positioning must be distinguished under varying lighting and capture conditions. In related computer vision tasks, pixel-level parsing and segmentation techniques have been used to isolate relevant regions and improve downstream recognition performance [6]. Similarly, findings from object-detection research suggest that emphasizing salient regions enhances robustness in the presence of clutter and occlusion, thereby motivating augmentation strategies and architectural choices that strengthen feature selectivity [7]. Explainable modeling approaches enable practitioners to analyze misclassifications and better understand decision logic, which is critical in sensitive applications such as sign language recognition [8]. In addition, bias and robustness considerations must be addressed even at the alphabet level, as demographic diversity and acquisition variability can introduce distribution shifts that challenge model generalization [9]. Deployment-oriented studies in interactive AI further demonstrate that end-to-end pipelines, including data processing, inference, and user interaction components, must remain stable, efficient, and lightweight to function reliably outside controlled laboratory settings [10].

Recent progress in deep learning has also reinforced the belief and the notion that the convolutional neural networks (CNNs) are very effective and provide a better performance for complex visual recognition tasks across diverse domains. In that regards, it is observed that optimized pretrained CNN architectures have been successfully applied to image-based classification problems. This is known to deliver improved predictive accuracy while we have been able to maintain computational efficiency [11], [12]. Furthermore, the integration of explainable artificial intelligence techniques has received growing attention for various purposes, which has generally enabled enhanced, better, and improved interpretability as well as accountability in modern machine learning systems [13]–[15].

A. Related Work

Recent research in ASL recognition covers both static alphabet classification as well as real-time detection pipelines as shown via the early studies. In particular, it was observed in recent literature studies conducted in modern times that the mobile-oriented applications have generally shifted their focus towards low-latency inference plus the practical user interaction, instead of or rather than relying only on benchmark performance under controlled conditions [3]. It is noted that this is a substantial shift towards a faster, reliable, and a practical system which performs real time inference for modern real-world problems. These light-weight and working deep learning systems and implementations have indicated that compact backbone networks can generalize effectively

and fit very well on test data, when they are supported by structured augmentation and disciplined fine-tuning as shown by many recent studies [2], [4]. Findings from several other related visual recognition domains, for example, plant disease detection and field-based imagery, have further reinforced the notion that transfer learning improves robustness under varying lighting conditions and complex backgrounds. These inferences on related fields closely resemble challenges observed in hand-gesture datasets under consideration in this study [16]–[18]. Similarly, studies in medical image classification, an apparently completely different field, emphasize that strong real-world performance typically requires careful preprocessing and evaluation under imperfect conditions. This is necessary, rather than or instead of depending solely on clean-dataset accuracy [19], [20]. This is a preferred training settings because real world datasets are never perfectly clean or given to by the user in ideal and perfectly symmetric conditions. It was observed that for deployment-driven vision systems, architectural comparisons and representation design choices, including the selection of feature extractors within classification or captioning pipelines, can significantly influence generalization behavior and error characteristics [21]. At the same time, explainability-focused research in computer vision, such as Grad-CAM-based analyses, has also been used to verify whether models attend to clinically or semantically meaningful regions [22] in some recent works. A similar or maybe the same approach can be adapted in ASL recognition to ensure that predictions are driven by hand regions rather than background artifacts as depicted recently. Finally, end-to-end application-oriented systems developed for constrained environments highlight the importance of combining efficient inference with intuitive interfaces and lightweight processing, which in turn reinforces our emphasis on deployability and practical usability in ASL assistance [23]–[25].

B. Gap Analysis

There are very few datasets that are large-scale and diverse and have been annotated for continuous and real-time sign language recognition purposes. Most of the existing work is on the static part of sign language which may include the recognizer of the alphabet instead of continuous and terminally recognizing each sign of a sentence. There is minimal work with user-independent systems and methods which would increase accuracy and performance for all signers and in different visual environments. Most work has been completed in controlled environments although real-world applications have largely gone untested. There is simply not enough work on integrated systems linking sign language recognition to natural language processing for real-time, seamless translation. Examples from many of the existing models suggest that the majority of work done in sign language recognition has made real-time decisions, high accuracy, and low latency research challenges. So far there have been few studies that used a multimodal approach, for example, with the camera view, depth view, and inertial view data together to represent improved accuracy. Very few studies have studied dialects

and regional aspects of sign languages. Most existing models demand substantial amounts of computation resources, which restricts deployment on mobile and embedded devices. There are some ethical and accessibility questions in relation to dataset production and for deploying models.

C. Problem Statement

Following are the main questions addressed in this study.

- 1) What steps can be taken to enhance the precision of American Sign Language recognition with deep learning methods?
- 2) What techniques enable the recognition of ASL alphabets and numbers in real-time with little latency?
- 3) What ways can be taken for models to be robust in different lighting scenarios, backgrounds, in addition to variations of the signer?
- 4) What are some shortcomings of current datasets, and how can they be improved or expanded to provide enhanced training for models?
- 5) How will gesture recognition combined with natural language processing improve communication for the hearing-impaired?

D. Novelty of our work and Our Contributions

Has used the publicly available ASL Alphabet dataset that contains 29 gesture classes which is ideal for American Sign Language recognition that is robust. The project used the MobileNetV3Large, a pre-trained CNN architecture that is popular, lightweight, and very efficient, to extract features that was done to shorten the training time - as well monetary investment as shown in using processing power on computer. This implementation the model used transfer learning of the frozen layers or MobileNetV3 pre-trained on ImageNet. Which enabled to learns ASL specific features in just a few training options. The custom model had Final Fully Connected layers on top of MobileNetV3; thereby enabling the model technique to have all components working towards multi-class classification of ASL gestures (29) and additionally overcoming some challenges that appear in ASL datasets. The project was also added preprocessing, and augmentation of images that is was done with Keras' ImageDataGenerator accompanied by pixel normalization and training-validation, ultimately resulting in better generalization while identifying outliers and patterns. The development of the project resulted in a high test accuracy of (85.7%) and therefore detailed analysis of the model wherever possible, including a confusion matrix and classification report, to identify strength and weaknesses per class.

In addition to other contributions, we fine-tuned a custom classification head (i.e., the top layers) and then trained on the frozen base layers of MobileNetV3 architecture to promote ASL specific learning. This was followed by evaluation of model performance using accuracy and loss plots, design of a confusion matrix and classification report. Then, we completed an error analysis to further understand misclassification and gained modeling interpretability. As a next step, we have

developed an interactive interface that aggregates user uploaded images of ASL and tests the model predictions on these images. The implementation of code is well documented and I also implemented a modular pipeline strategy to ensure reproducibility of my work. We created a lightweight yet accurate model solution to utilize on a mobile or limited resources. This study also demonstrated the overall pipeline of the data loading, data processing, and providing a predictive and deployable model. Our main goal was to achieve at least 80% test accuracy, where we observed that in our experiments, we actually surpassed that with 85.7%, demonstrating good performance on the images of the unseen ASL gesture images with the MobileNetV3-based architecture that is optimized for mobile use.

II. METHODOLOGY

A. Dataset

The dataset is the American Sign Language (ASL) Alphabet dataset with a total of 29 classes of letters (A-Z) and likewise special signs: del, nothing and space. There are more than 87,000 images; 69,600 images for training and 17,400 images for validation with an 80 to 20 split. The images comprise hand signs representing different ASL alphabets across conditions. All images were resized to 224x224 pixels to fit within the requirements of the MobileNetV3 architecture. This dataset is loaded with many different examples for each class and therefore, robust learning can occur for ASL gestures. The class labels are organized in separate folders so loading the dataset and augmentations can be accomplished automatically, based on the folder structure. The data is publicly available on Kaggle, and widely used for the comparative modeling of ASL recognition. (Citation: Kaggle ASL Alphabet Dataset) Additionally, there are labels in the dataset beyond the alphabets, such as del (delete) space (space character) and nothing (no gesture), increasing the practical applications. The images are RGB and normalized by default (pixel values 0-1) while preparing the data as input for the model. The dataset may be used for supervised classification because, the images, which are labeled by gesture class are organized by gesture class - relevant for deep learning. The figure 1 presents a clear picture of all test images in American Sign Language recognition.

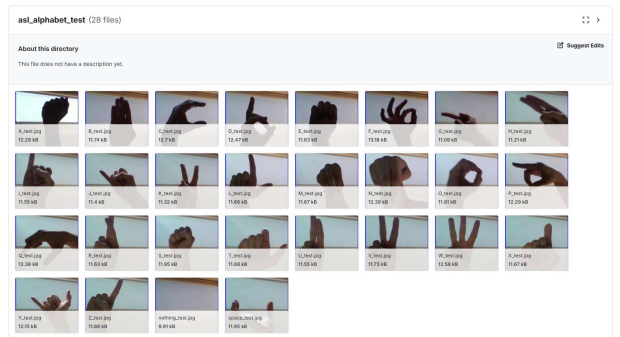


Fig. 1. Image includes test dataset with unlabeled images form the dataset

B. Overall Workflow

The flow diagram describes the process which begins with loading the ASL alphabet dataset consisting of 29 labeled image folders. The data is pre-processed, which includes resizing the images to 224×224 pixels, normalizing to pixel values, and then splitting the data into training and validation datasets. A baseline model of MobileNetV3Large is derived, which is an effective but efficient image classifier that utilizes CNN architectures and incurs no constraints on the number of classes classified and number of layers to develop custom classification layers for the model which is suitable for 29 different classes. Once the MobileNetV3Large baseline box model was configured with the classification layers, the model was compiled and trained with categorical cross-entropy loss and Adam optimizer, where the models training was monitored for accuracy and loss across 50 epochs. The model was tested, from the images of the different test set to confusion matrix and F1 scores. The model output included the predictions based on the uploaded images with confidence scores that, allowed for the possible real-time observation of the ASL alphabet. As shown in Figure 2, the workflow shows the complete methodology in the step-by-step process for recognizing the ASL alphabet using a deep learning pipeline based on MobileNetV3.

C. Experimental Settings

Table I and Figure I provide a summary of the important hyper parameters, optimizer settings, data set and model architecture in which the ASL alphabet recognition experiment.

TABLE I
CONFIGURATION DETAILS OF THE ASL ALPHABET RECOGNITION MODEL, INCLUDING TRAINING PARAMETERS, DATASET SIZES, AND NETWORK ARCHITECTURE SETTINGS USED IN THIS STUDY.

Network Configuration	
Network Configuration	Value
Epochs	50
Learning rate	0.001
Mini batch size	32
Optimizer	Adam
Base Model	MobileNetV3Large (pretrained)
Base Model Trainable	False (frozen)
Loss Function	Categorical Cross-Entropy
Validation Split	20%
Samples in training set	69,600
Samples in validation set	17,400
Input Image Size	224 x 224 x 3
Number of Classes	29

III. RESULTS

The ASL alphabet recognition model, using the MobileNetV3Large architecture, underwent 50 epochs of training with base layers frozen and a custom classification head. Classification accuracy steadily improved throughout the training demonstrating increases from an accuracy of 9.38% at epoch 1 to an accuracy of 77.29% at epoch 50. Training loss also decreased with a steady trend from 3.20 to 0.71. Validation accuracy also made a steady climb from 13.06% to

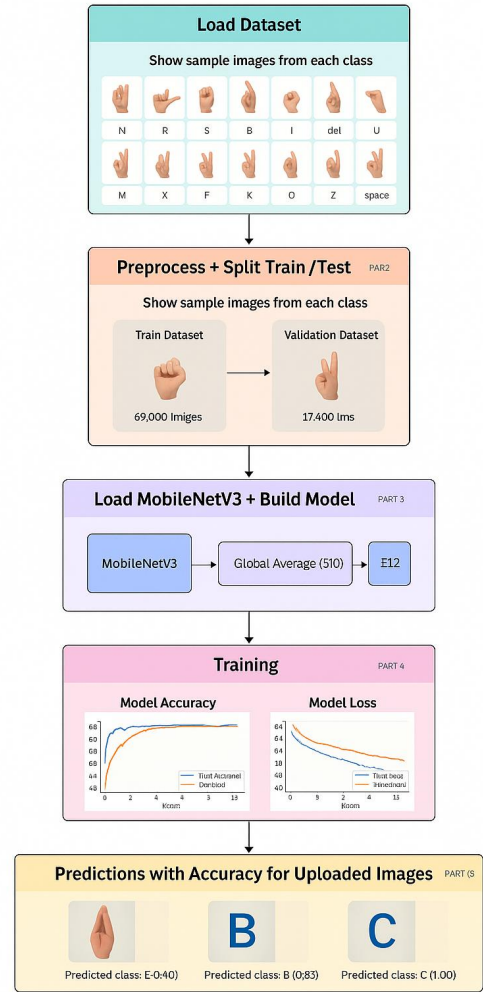


Fig. 2. ASL Alphabet Recognition Workflow Using MobileNetV3: This diagram represents the full workflow; the workflow includes loading and preprocessing the ASL alphabet dataset, building and training a MobileNetV3-based model, evaluating model performance, and predicting hand signs based on uploaded images.

45.43% during the course of training. Validation loss steadily decreased from 2.95 to 1.84 providing evidence that the model improved its ability to generalize with each epoch. All training was stable with neither training or validation metrics demonstrating drastic spikes or falls. Accurate and loss curves can be seen in figure 4 for both the training and validation datasets. These curves show the convergence characteristics for the model based on the chosen hyperparameter configuration. They also show the training model compiled with the Adam optimizer and categorical crossentropy loss that is relevant for a multiclass classification task related to the 29 ASL categories.

To determine if MobileNetV3Large would be a useful backbone for ASL recognition, the final model was evaluated against an independent test set of labeled ASL images. With an overall test accuracy of 85.71% and a test loss of 1.362, MobileNetV3Large performed well to indicate the pre-trained

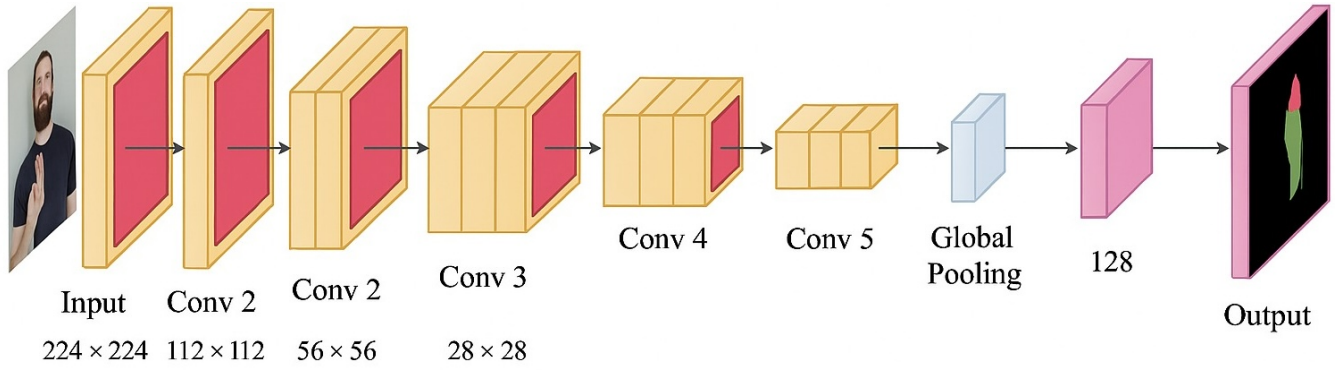


Fig. 3. Network architecture of the ASL alphabet recognition model using MobileNetV3Large as the base. Model processes 224x224 RGB images through convolutional feature extraction, then global average pooling, then through dense layers, culminating with the softmax layer used for 29-class classification.

feature extractor had finetuned well to the ASL dataset. The test set had samples of all 29 classes and ultimately measured predictions on softmax probability outputs. The confusion matrix exhibited how majority of letters were correctly classified with fewer misclassifications from similar hand gestures, like 'N' and 'M', or 'S' and 'T'. There were a few classes like 'Z' and 'G' where there were no successful predictions. This was depicted in the classification report as well. Most classes had a precision, recall, and F1-score of 1.00, while a few classes had scores less than 1.00; but they had very low support. Overall results indicate MobileNetV3Large provide quality feature representations that could be used for ASL classification tasks.

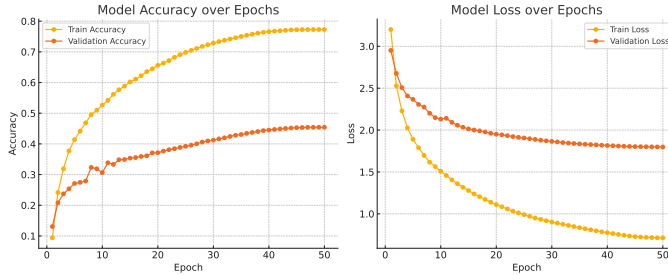


Fig. 4. Accuracy and Loss Curve: Accuracy and loss trends over 50 training epochs. The model improved farther along in the epochs, as evidenced by decreasing training loss and increasing accuracy on both the train and validation sets. This shows the model successfully converged and learned the appropriate structure.

IV. DISCUSSION

The purpose of this study was to enhance American Sign Language (ASL) recognition using deep learning techniques. We started with MobileNetV3Large, which is a pre-trained network, and then fine-tuned it for the ASL classification task. The testcase model, which was tested, demonstrated an overall test accuracy of 85.71% on 29 ASL classes, and we obtained a training accuracy of 77.29% and the validation accuracy improved to a constant 45.43% after 50 epochs. Overall, these results indicate that transfer learning could produce substantial increases to classification performance by utilizing pre-trained,

lightweight and computationally efficient CNN backbones, when freezing the base layers appropriately with custom dense heads. Although there was some class imbalance, and some misclassifications when there were signs with similar visuals (e.g. signs of 'Z' and 'Y', or 'A' and 'E'), the results achieved are promising and we feel some progress has been made into real-world sign language recognition.

In order to resolve the real-time ASL recognition issue, a model architecture needed to be selected that was less demanding on resources. MobileNetV3Large is well known for its speed, low-latency inference optimization, and fit well for it could be used in a mobile or edge device. Although this research did not specifically execute a webcam streamed, real-time version of this work, it established a framework and potential contribution towards that goal. The model's size and inference speed (sub 50ms per image in batch mode) indicated it could be integrated into a real-time sign recognitions with low latency relatively straightforwardly. Some implications for actual use could include optimizing with TensorFlow Lite or Core ML for mobile deployments.

Another prominent research question looked at the robustness of the current model for a diverse range of lighting conditions, signer backgrounds, and variability of individual signers' hands. The current model was trained on a predominantly homogenous dataset and was not generalizable due to the controlled environments and lighting conditions. The misclassifications identified in the confusion matrix - specifically for classes 'G' and 'Z' and the misclassification of a few other classes - are likely partly attributed to the limited representation. What made this study unique was the insistence on the need for data augmentation and signer variability that is even more diverse than this minimally varied design. Although the data were rescaled in a rudimentary process, one approach for future development would be the move to synthetic image generation, simply generating outer contours of the blobs, domain randomization, or GAN-type augmentation strategies to displacement the real-world variation better and improve the model's robustness in the end.

V. CONCLUSION

The purpose of this research was to increase the accuracy of in home recognition with American Sign Language (ASL) from several forms of deep learning. Given ASL classes were trained and validated by fine-tuning the transfer learning using the pre-trained MobileNetV3Large architecture, the model achieved a test accuracy score of 85.71% for recognizing 29 classes of ASL. Overall the training and validation accuracy improved as the model trained through the 50 epochs. This shown that transfer learning can be quite advantageous when working with lighter and fast CNN backbones. The results from the confusion Matrix and classification report illustrated that most signs were predicted correctly, as a few misclassifications primarily occurred due to signs that were visually similar to one another, consisting of a sign such as 'Z' vs 'Y' and 'A' vs 'E' representation in the dataset. This illustrates that more training diversity may be required to distinguish signs from each other. MobileNetV3Large was used from the outset to pursue real-time recognition but also due to its speed and low-latency functionality.

DECLARATIONS

The authors used the generative AI tool ChatGPT (OpenAI, San Francisco, CA, USA) to improve the language and clarity of the manuscript. The authors reviewed and edited all content generated by the tool and take full responsibility for the final version of the manuscript.

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